

Research Interests

- **Computational / mathematical linguistics:** grammatical inference; finite-state modeling; theory-driven, interpretable, data-efficient ML; low-resource corpus design; evaluation design
- **Phonology:** formal modeling of morphophonology; learnability; typology; constructing grammars for understudied languages; tone
- **Natural language understanding:** cognitive states, common ground, Theory of Mind; annotation/corpus design for mental-state inference; linguistically informed features; structured representations in modern NLP/LLMs; rigorous evaluation

Current Position

Sep 2026 **Postdoctoral Research Fellow**, *University of Surrey*, REVOLUPHON Project
Working on computational and theoretical approaches to phonology as part of the REVOLUPHON research project.

Past Projects

- **Learnability of Morpho-Phonological Grammars**
 - Developing interpretable learners for morphophonological mappings that learn effectively from limited data by encoding constraints from subregular phonology and linguistically structured representations.
 - Supervised by Dr. Jeffrey Heinz.
 - Partially funded by NSF Award No. 2416184
- **The Common Ground Corpus**
 - Creating a corpus for computational pragmatics by annotating unscripted dialogs from the CALLHOME corpus with common ground information from the perspective of the dialogs' participants.
 - Leading a team of undergraduate and graduate students with Dr. Owen Rambow.
 - Partially funded by DARPA under No. HR001120C0037 and PR No. HR0011154158.
- **Documentation and Computational Modeling of Shupamem**
 - Conducting linguistic field research and analysis of Shupamem (Grassfields Bantu).
 - Developing finite-state models of Shupamem's morpho-phonology with special attention to reduplication and tone.

Education

June 2026 **PhD in Linguistics**, *Stony Brook University*, Thesis: *"Learning Phonological Transductions From Sparse Data Over Structured Representations"*
Dissertation defended: March 13, 2026
Advisor: Dr. Jeffrey Heinz

2020 Sep **MA in Computational Linguistics**, *The Graduate Center, CUNY*; Thesis: *"Tones In Shupamem Reduplication"*
Advisor: Dr. Kyle Gorman

2013 Jun **MA in English Philology**, *University of Warsaw*; Thesis: *"An Optimality Theoretic analysis of contextual and contrastive nasalization in English and Polish"*
Advisor: Dr. Bartłomiej Czaplicki

2011 Jun **BA in English Philology**, *University of Warsaw*

Publications

- 2026 July **Magdalena Markowska** and Jeffrey Heinz. *Learning k-ISL functions with phonological features*. Journal of Language Modelling 14.1: 1-80
- 2024 August Adil Soubki, John Murzaku, Arash Yousefi Jordehi, Peter Zeng, **Magdalena Markowska**, Seyed Abolghasem Mirroshandel, Owen Rambow. *Views Are My Own, but Also Yours: Benchmarking Theory of Mind Using Common Ground*. Findings of the Association for Computational Linguistics.
- Jan Hagay Schurr, Jason Kandybowicz, Abdoulaye Laziz Nchare, Tysean Bucknor, Xiaomeng Ma, **Magdalena Markowska**, and Armando Tapia. *Absence of Clausal Islands in Shupamem*. Languages 9.7: 1–35.
- 2023 Dec **Magdalena Markowska**, Mohammad Taghizadeh, Adil Soubki, Seyedabolghasem Mirroshandel, Owen Rambow *Finding Common Ground: Annotating and Predicting Common Ground in Spoken Conversations*. Findings of Empirical Methods in Natural Language Processing.
- Jun **Magdalena Markowska** and Jeffrey Heinz. *Empirical and Theoretical Arguments for Using Properties of Letters for the Learning of Sequential Functions*. Proceedings of Machine Learning Research 1:1–5.
- 2022 Sep John Murzaku, Peter Zeng, **Magdalena Markowska**, and Owen Rambow. *Predicting the Author's Presentation of Factuality*. Proceedings of the 29th International Conference on Computational Linguistics.
- Jul Jordan Kodner, Salam Khalifa, Khuyagbaatar Batsuren, Hossep Dolatian, Ryan Cotterell, Faruk Akkus, Antonios Anastasopoulos, Taras Andrushko, Aryaman Arora, Nona Atanalov, Gábor Bella, Elena Budianskaya, Yustinus Ghanggo Ate, Omer Goldman, David Guriel, Simon Guriel, Silvia Guriel-Agiashvili, Witold Kieraś, Andrew Krizhanovsky, Natalia Krizhanovsky, Igor Marchenko, **Magdalena Markowska**, Polina Mashkovtseva, Maria Nepomniashchaya, Daria Rodionova, Karina Scheifer, Alexandra Sorova, Anastasia Yemelina, Jeremiah Young, Ekaterina Vylomova. 2022. *SIGMORPHON–UniMorph 2022 Shared Task 0: Generalization and Typologically Diverse Morphological Inflection*. In Proceedings of the 19th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology, pages 176–203, Seattle, Washington. Association for Computational Linguistics.
- Jun Jennifer Tracey, Owen Rambow, Claire Cardie, Adam Dalton, Hoa Trang Dang, Mona Diab, Bonnie Dorr, Louise Guthrie, **Magdalena Markowska**, Smaranda Muresan, Vinodkumar Prabhakaran, Samira Shaikh and Tomek Strzalkowski. *BeSt: The Belief and Sentiment Corpus*. Proceedings of the 13th Edition of Language Resources and Evaluation Conference.
- 2021 Aug **Magdalena Markowska**, Jeffrey Heinz, and Owen Rambow. *Finite-state Model of Shupamem Reduplication*. Proceedings of the 18th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology.
- Qihui Xu, **Magdalena Markowska**, Martin Chodorow, and Ping Li. *Modeling Bilingual Lexical Processing through Code-Switching Speech: A Network Science Approach*. Frontiers in Psychology.
- Feb Qihui Xu, **Magdalena Markowska**, Martin Chodorow, and Ping Li. *A Network Science Approach to Bilingual Code-switching*, Proceedings of the Society for Computation in Linguistics: Vol. 4 , Article 3. DOI: <https://doi.org/10.7275/raze-1b18>.
- 2019 Nov Kyle Gorman, Arya D. McCarthy, Ryan Cotterell, Ekaterina Vylomova, MiikkaSilfverberg, and **Magdalena Markowska**. *Weird inflects but OK: making sense of morphological generation errors*. Proceedings of Conference on Computational Natural Language Learning, 140-151.

Relevant Experience

Research

2025 Summer **Learning Underlying Representations in Phonology**, *Research Assistant*, PI: Dr. Jeffrey Heinz

2023 Spring **Cognitive states modelling**, *Research Assistant*, PI: Dr. Owen Rambow

2021-22 **Cognitive states modelling**, *Research Assistant*, PI: Dr. Owen Rambow

Teaching: Primary Instructor

2025 Summer **Structure of English (LIN 320/527)**, *Primary Instructor*, Stony Brook University

2025 Winter **Languages in the US (LIN 200)**, *Primary Instructor*, Stony Brook University

2024 Fall **Structure of English (LIN 320/527)**, *Primary Instructor*, Stony Brook University

2023 Summer **Structure of English (LIN 320/527)**, *Primary Instructor*, Stony Brook University

Teaching: Teaching Assistant

2025 Fall **Phonology (LIN 301)**, *Teaching Assistant*, Stony Brook University

2025 Fall **Historical Linguistics (LIN 340)**, *Teaching Assistant*, Stony Brook University

2025 Spring **Computational Linguistics (LIN 335)**, *Teaching Assistant*, Stony Brook University

2024 Spring **Languages in the US (LIN 200)**, *Teaching Assistant*, Stony Brook University

2023 Fall **Language and Technology (LIN 120)**, *Teaching Assistant with recitations*, Stony Brook University

2021 Spring **Survey in Cognition and Perception (PSY 260)**, *Teaching Assistant*, Stony Brook University

2020 Fall **The Structure of an Uncommonly Taught Language (LIN 431)**, *Teaching Assistant with recitations*, Stony Brook University

Talks

2025 Nov **Magdalena Markowska**, *Efficient learning of phonological generalizations with String Equation Adaptive Learner (SEAL)*, The 2025 Rutgers Subregular Morphophonology Workshop

May **Magdalena Markowska**, *Learning from small data: features, locality and directionality*, Invited talk at North American Phonology Conference (NAPhC)

April **Magdalena Markowska**, *Learning from small data: features, locality and directionality*, PhPhoNE Workshop at MIT

2024 Nov **Magdalena Markowska and Jeffrey Heinz**, *Learning k -ISL functions over phonological features: case studies*, Poster at the Annual Meeting on Phonology (AMP)

Sep **Magdalena Markowska**, *Learning ISL functions with features*, Rutgers Subregular Phonology Workshop

2023 Sep **Magdalena Markowska, Adil Soubki, Gary Mar, Sayed Mirroshandel, Owen Rambow and Anita Wasilewska**, *Formal Representation of Common Ground in Dialogue*, Fourteenth International Tbilisi Symposium on Logic, Language and Computation (TbiLLC 2023)

Jul **Magdalena Markowska**, *Empirical and Theoretical Arguments for Using Properties of Letters for the Learning of Sequential Functions*, International Conference on Grammatical Inference (ICGI 16)

Apr **Magdalena Markowska**, *Learning phonological transformations with small data using sub-symbolic properties*, IACS Student Seminar

2022 Apr **Magdalena Markowska**, *The Common Ground Corpus*, IACS Student Seminar

- Magdalena Markowska**, *Tonal Polarity and Compound Nouns in Shupamem*, Annual Conference on African Linguistics (ACAL 53)
- 2021 Oct **Magdalena Markowska**, *Shupamem DP: tonal changes on nouns modified by adjectives*, Grassfields Bantu Working Group at McGill University
- Aug **Magdalena Markowska**, *Finite-state Model of Shupamem Reduplication*, SIGMORPHON 2021: 18th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology
- Mar **Magdalena Markowska**, *Finite-state Analysis of Tonal Reduplication in Shupamem*, IACS Student Seminar, Stony Brook University
- Magdalena Markowska**, *Tones in Shupamem Reduplication*, CIRCLE, The Graduate Center, CUNY

Awards

- 2025 **Stony Brook Foundation Board of Trustees Dissertation Completion Endowed Fellowship**, SBU
- 2022,2023 **Junior Researcher Award**, *Institute for Advanced Computational Science, SBU*
- 2022 Jul **Full scholarship to attend the African Linguistics School (ALS 6)**, *Porto Novo, Benin*
- 2020,2021 **New Recruit Award**, *Institute for Advanced Computational Science, SBU*

Service

- 2025 **Reviewer**, OCP23 Workshop on The Role of Representation in Computational Phonology
- 2023 Dec - present **Reviewer**, Association for Computational Linguistics Rolling Review
- 2022-4 **Secretary and Vice President, IACS Student Association**, *Institute for Advanced Computational Science, Stony Brook University*
- 2021-2 **Social Media Manager**, *Linguistics Department, Stony Brook University*

Certificates

- 2024 **Advanced Graduate Certificate in Data and Computational Science**, *Institute for Advanced Computational Science, Stony Brook University*

Technical Skills

- **ML (general)**: PyTorch; scikit-learn; experiment tracking; evaluation & error analysis
- **Algorithms and Formal Methods**: finite-state modeling (Pynini/OpenFst); grammatical inference; formal language theory
- **NLP (core workflows)**: corpus construction & dataset curation; tokenization; classical NLP (spaCy, NLTK); feature engineering; regex
- **Programming & Data**: Python; pandas/NumPy; SQL; Git; Jupyter; LaTeX

Natural Languages

Polish (native), English (advanced), German (pre-intermediate), French (pre-intermediate), Shupamem (fieldwork)